

Run 11 | OME

India

IV + controls

Target Opportunity-motivated entrepreneurship; binary 1 vs. 0
Sample 27,555 respondents **Coverage** 1 country; 2015–2022
Positive cases 2,167 (7.86%) **Case time** 57s
Train / val. / test 18,739 / 3,306 / 5,510 (68% / 12% / 20%, stratified)
Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.
Fraser EFW predictors (7): Overall economic freedom, Economic freedom quartile, Size of government, Legal system and property rights, Sound money, Freedom to trade internationally, Regulation. Total independent variables: 20.
Controls (10): Survey year, Age, Gender, Education attainment, Household income, Household size, Occupational status, GDP per capita (PPP), Population, GDP growth. Country and respondent ID are excluded as predictors.
Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: Random Forest (0.9876). Highest F1: Ridge (0.7910). For Random Forest, adding controls changed ROC-AUC by +0.0004 on the same test records.

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
Random Forest	0.9876	0.8557	96.12	71.02	85.45	0.7757	1s
H2O GBM	0.9875	0.8508	96.21	73.73	80.37	0.7691	3s
CatBoost	0.9872	0.8481	96.06	72.22	81.06	0.7639	5s
XGBoost	0.9872	0.8442	96.15	72.78	81.52	0.7691	<1s
Lasso	0.9868	0.8449	96.23	70.79	88.45	0.7864	12s
Ridge	0.9868	0.8457	96.28	70.80	89.61	0.7910	11s
Logistic Regression	0.9867	0.8441	96.24	70.40	90.07	0.7903	2s
Elastic Net	0.9865	0.8433	96.17	71.94	84.06	0.7753	14s
Decision Tree	0.9845	0.7893	95.63	67.78	84.53	0.7523	<1s
Naive Bayes	0.9126	0.5739	92.94	54.21	65.36	0.5927	6s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	763.76	40.38
2	Established business owner	272.64	14.42
3	Business angel	78.01	4.12
4	Occupational status	73.83	3.90
5	Knowing entrepreneurs	66.88	3.54
6	Age	64.51	3.41
7	Household size	47.41	2.51
8	Business exit	44.96	2.38

Share is the percentage of total forest impurity importance across all 53 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Fraser EFW predictors within the same forest

Rank	Fraser EFW predictor	Raw importance	Share%
1	Sound money	17.55	0.93
2	Regulation	15.41	0.81
3	Freedom to trade internationally	14.78	0.78

Fraser EFW subset ranked by raw importance; shares use total forest importance, not the Fraser EFW subtotal.

Source: Completed Objective 3 batch, 29 September 2026; 11_obj3_india_iv_cv_OME. Scores and importance are taken directly from saved CSV outputs.